PROJECT PLANNING (PA2-2026)

1. Introduction

Assignee	Reviewer	Editor
Nghĩa	Văn	Chien

1.1 Purpose

This Project Planning document describes the overall approach, schedule, and management strategy for developing **LoveYou**, an AI-enhanced dating web application designed to help Vietnamese users find meaningful romantic connections online. It serves as the roadmap for the team during development and is used to coordinate work, track progress, and control delivery.

1.2 Scope

This plan covers the development of a responsive web application with both frontend and backend components. The project includes the implementation of core AI features such as Smart Matching, Red Flag Detection, and an Ice-Breaking game, as well as support for user profiles, avatar and photo uploads, chat history, and structured database management.

1.3 Overview

This document is organized into the following parts: Project Overview, Project Organization, Project Plan, and Project Monitoring and Control. It also includes sprint planning, build planning, team roles, assumptions, deliverables, and risk management. The plan is written in Markdown format as required by PA2.

2. Project Overview

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Nghĩa	Hoang	Chien

2.1 Project Purpose, Scope, and Objectives

The purpose of LoveYou is to provide a secure, user-friendly, and intelligent dating platform for Vietnamese users. The project aims to improve online matchmaking quality through AI support, help users communicate more effectively, and provide a modern experience across desktop and mobile devices.

The main objectives are:

- Provide a responsive and accessible dating web application.
- Implement AI-assisted matching and red-flag detection.
- Support user profiles, photos, interest tags, and chat features.
- Deliver a stable full-stack system with proper database integration.
- Complete all course deliverables on time using the Scrum process.

2.2 Assumptions and Constraints

Assumptions

- Team members are available throughout the semester and can coordinate through Discord, Trello, and GitHub.
- Internet access is available for development and testing.
- External AI tools, cloud hosting services, and GitHub remain accessible during development.
- Team members will complete the required Spec Kit training and self-learning activities.

Constraints

- The project must be completed within the course timeline.
- Development is limited to a student team and free or low-cost tools.
- All reports and submissions must follow the course requirements.
- The final solution must be feasible within the available time and technical resources.

2.3 Project Deliverables

Deliverable	Planned Sprint
Project Proposal	PA1
Project Plan (initial draft)	PA2
Vision Document (initial draft)	PA2
Spec Kit initialization artifacts	PA2
Revised Project Plan	PA3
Revised Vision Document	PA3
Use-Case Models and Specifications	PA3
UI Prototype	PA3
Software Architecture Document	PA4
Additional functional builds	PA3 - PA5
Test Plan and Test Cases	PA5
Final source code	PA5
Final demo and reflective report	PA5

3. Project Organization

Assignee	Reviewer	Editor
Nghĩa	Huy	Van

3.1 Team Structure and Roles

ID	Full Name	Role
19127484	Ngô Trung Nghĩa	Team Leader, Database
23127471	Lê Văn Hoàng Tấn	UI/UX, Framework
23127331	Nguyễn Công Chiến	Developer
23127368	Nguyễn Minh Hoàng	Developer, Marketing Product
23127197	Nguyễn Tường Huy	Developer, Backend
20127095	Vũ Lê Trọng Văn	Tester, Technical Support

3.2 Roles and Responsibilities

Member	Responsibilities
Ngô Trung Nghĩa	Manage the project, coordinate sprint activities, handle database design, review deliverables, and compile submission files.
Lê Văn Hoàng Tấn	Design the UI/UX, support frontend framework decisions, and prepare the Vision Document content.
Nguyễn Công Chiến	Set up the repository, support Spec Kit initialization, and implement assigned development tasks.
Nguyễn Minh Hoàng	Support feature development, product analysis, and documentation tasks.
Nguyễn Tường Huy	Develop backend APIs, assist with integration, and support server-side logic.
Vũ Lê Trọng Văn	Test features, support technical tasks, and help with quality assurance and reporting.

3.3 Risk Management

Risk Description	Impact	Mitigation Strategy
Member unavailability	High. Sprint tasks may be delayed and deadlines may be missed.	Maintain clear documentation in Discord/GitHub, split tasks into smaller items, and let other team members temporarily cover critical work when needed.
Technology issues with AI	Medium. AI matching or red-flag features may fail or return inconsistent results.	Research AI APIs early, test integration in advance, and prepare a non-AI fallback matching method to keep the application usable.
Scope creep	High. Extra features may prevent the team from finishing the core product on time.	Follow the predefined functional groups, require team consensus for new features, and move non-essential items to the backlog.

4. Project Plan

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Nghĩa	Tan	Van

The project follows the Scrum framework and is organized into five sprints, each corresponding to one course assignment. Scrum meetings are held regularly every Friday evening or during the weekend, and progress is tracked using Trello/Jira.

4.1 Project Estimates

The project is expected to be completed within five sprints. The estimate below is based on the current team size, the planned feature set, and the course timeline.

Activity	Estimated Effort
Planning and documentation	80 hours
UI/UX design	40 hours
Frontend development	120 hours
Backend development	120 hours
Database design and integration	40 hours
Testing and debugging	60 hours
Deployment and final polishing	20 hours
Total	**480 hours**

4.2 Sprint Plan

4.2.1 Sprint Overview

Sprint	Course Milestone	Timeline	Main Deliverables
Sprint 1	PA1	25/05/2026 – 07/06/2026	Project Proposal, Team Contract, Repository Setup
Sprint 2	PA2	08/06/2026 – 12/07/2026	Project Plan, Vision Document, Spec Kit, AI Usage Report
Sprint 3	PA3	13/07/2026 – 25/07/2026	Revised Project Plan, Revised Vision, Use Case Model, UI Prototype, Build 1
Sprint 4	PA4	26/07/2026 – 09/08/2026	Software Architecture Document, Build 2
Sprint 5	PA5	10/08/2026 – 23/08/2026	Final Testing, Build 3, Final Demo, Final Submission

4.2.2 Sprint Objectives and Tasks

Sprint 1: PA1 (Project proposal, team contract, initial feature list, and setup)

- **Timeline:** 08/06/2026 - 21/06/2026
- **Objectives:** Finalize project idea and scope. Complete the project proposal and existing app survey. Set up the team contract and development environment.
- **Tasks:**
  - T1: Finalize project idea and scope.
  - T2: Complete the project proposal and existing app survey.
  - T3: Set up the team contract and development environment.

Sprint 2: PA2 (Initial project plan, vision document, Spec Kit initialization)

- **Timeline:** 22/06/2026 - 06/07/2026
- **Objectives:** Draft the initial Project Plan and Vision Document. Initialize Spec Kit in the GitHub repository. Complete the AI usage report and weekly report materials. Collect evidence of Spec Kit training.
- **Tasks:**
  - T1: Draft the initial Project Plan and Vision Document.
  - T2: Initialize Spec Kit in the GitHub repository.
  - T3: Complete the AI usage report and weekly report materials.
  - T4: Collect evidence of Spec Kit training.
  - T5: Plan overall architecture and define database schema.

Sprint 3: PA3 (Use-case model, UI prototype, and Build 1)

- **Timeline:** 07/07/2026 - 25/07/2026
- **Objectives:** Revise the Project Plan and Vision Document based on TA feedback. Create use-case diagrams, specifications, and UI prototypes. Implement one complete full-stack functional group.
- **Tasks:**
  - T1: Revise the Project Plan and Vision Document based on TA feedback.
  - T2: Create use-case diagrams, specifications, and UI prototypes using Figma.
  - T3: Create and handle API endpoints for the first functional group.
  - T4: Implement frontend UI and connect with backend for the first functional group.

Sprint 4: PA4 (Architecture documentation and Build 2)

- **Timeline:** 26/07/2026 - 09/08/2026
- **Objectives:** Document software architecture using C4 diagrams and a deployment diagram. Implement two additional functional groups. Improve integration, usability, and code quality.
- **Tasks:**
  - T1: Document software architecture (C4 diagrams) and deployment diagram.
  - T2: Implement AI Smart Matching and Chat APIs on the backend.
  - T3: Implement frontend UI for Matching and Chat features.
  - T4: Integrate frontend with backend and test two additional functional groups.

Sprint 5: PA5 (Testing, final build, final demo, submission)

- **Timeline:** 10/08/2026 - 24/08/2026

- **Objectives:** Complete testing and bug fixing. Implement remaining functional groups. Prepare the final build, live demo, and reflective report.
- **Tasks:**
  - T1: Write unit tests, test full user flows, and fix bugs.
  - T2: Implement remaining features (Red Flag Detection, Ice-Breaking game).
  - T3: Refactor code, optimize performance, and prepare the final build.
  - T4: Present and demo the application, submit the final report and deliverables.

4.2.3 Releases

Release	Description
Build 1	End of PA3. First functional build covering one complete functional group implemented end-to-end.
Build 2	End of PA4. Beta build integrating two additional functional groups and validating system integration.
Build 3	End of PA5. Final release with all features implemented, tested, and polished for the demo.

4.2.4 Project Schedule

Sprint	Start Date	End Date	Major Deliverables
Sprint 1	25/05/2026	07/06/2026	Project Proposal, Team Contract
Sprint 2	08/06/2026	12/07/2026	Project Plan, Vision Document, Spec Kit, AI Usage Report
Sprint 3	13/07/2026	25/07/2026	Revised Documents, Use Case Model, UI Prototype, Build 1
Sprint 4	26/07/2026	09/08/2026	Software Architecture Document, Build 2
Sprint 5	10/08/2026	23/08/2026	System Testing, Final Build, Demo, Final Submission

4.2.5 Project Resourcing

The project team consists of six members and uses the following tools and technologies:

- ReactJS for the frontend
- NodeJS and ExpressJS for the backend
- MongoDB for the database
- GitHub for version control and collaboration
- Trello or Jira for task tracking
- Spec Kit for structured specification development
- AI APIs for matching and red-flag support

Team members will complete self-training on these technologies during Sprint 2 and will continue improving their skills throughout the project.

4.3 Project Monitoring and Control

4.3.1 Requirements Management

Requirements are maintained in the Vision Document and tracked through the team task board. Any change to scope or feature behavior must be discussed in sprint meetings and approved by the team leader before implementation.

4.3.2 Reporting and Measurement

Progress will be reviewed through weekly Scrum meetings and sprint reviews. The team will track:

- Completed tasks
- Remaining tasks
- Sprint velocity
- Number of open issues
- Number of completed features

4.3.3 Risk Management

Project risks will be reviewed at least once per sprint. The team will update the risk table when new issues appear and will adjust mitigation plans as necessary.

4.3.4 Configuration Management

All source code and documentation will be stored in GitHub with version control enabled. Pull requests and review comments will be used to manage changes before merging into the main branch. Important deliverables and artifacts will be tagged and archived for final submission.

5. Build Plan

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The team plans to produce three major software builds throughout the project lifecycle. Each build will be used for testing, integration validation, and stakeholder review before moving to the next stage.

5.1 Build 1 (End of PA3)

Purpose:

- Validate the project architecture
- Demonstrate one complete functional group implemented end-to-end
- Verify frontend, backend, and database integration

Included Features:

- User authentication and account management
- Initial database integration
- Basic user interface framework

Testing Activities:

- Unit testing.
- Basic integration testing.
- User interface validation.

5.2 Build 2 (End of PA4)

Purpose:

- Validate communication among major system components
- Evaluate system stability before final implementation

Included Features:

- Build 1 features
- Two additional functional groups
- Initial AI feature integration
- Enhanced user profile management and matching functions

Testing Activities:

- Functional testing
- Integration testing
- Architecture validation
- Performance verification

5.3 Build 3 (End of PA5)

Purpose:

- Deliver the complete product for final demonstration and evaluation.

Included Features:

- All functional groups.
- Complete AI-assisted matching features.
- Red Flag Detection.
- Ice-Breaking Game.
- Full chat and profile management capabilities.

Testing Activities:

- System testing.
- Acceptance testing.
- Regression testing.
- Final bug fixing and UI polishing.